

# Telescopes & Surveys — Study Notes

CASSA RA (Astrophysics) 2026-27 · companion to the practice sheet

Why this matters for the viva: your two data sources are **SDSS-II** (Paper 1 hosts) and **CSP-I** (Paper 2). Panel may ask "where does your data come from?", "why 7 bands?", "what is LSST?" Know Sections 1-3 cold; skim 4-6 for breadth.

## SECTION 1 — Your two core surveys (know in depth)

### SDSS — Sloan Digital Sky Survey

- **Telescope:** dedicated **2.5 m** at **Apache Point Observatory (APO)**, New Mexico, USA.
- **Filters:** **u g r i z** (5 optical bands, the "SDSS system",  $\sim 3500\text{-}9000\text{ \AA}$ ).
- **SDSS-II Supernova Survey (2005-2007):** repeatedly scanned **Stripe 82** (equatorial strip) →  **$\sim 500$  spectroscopically confirmed SN Ia** at **intermediate redshift  $z \approx 0.05\text{-}0.4$** . This filled the gap between local and high- $z$  SN on the Hubble diagram.
- **Phases:** SDSS-I/II (2000-08) → III (BOSS) → IV (eBOSS, **MaNGA** IFU, APOGEE) → **SDSS-V** (ongoing: Milky Way / Black Hole / Local Volume Mappers).
- **Your link:** the **Rose et al. (2019)** FSPS SED-fitting pipeline you rebuild uses **SDSS-II host galaxies** (ugriz host photometry) — that is your Paper 1 sample.

### CSP — Carnegie Supernova Project

- **Observatory:** **Las Campanas**, Chile.
- **Telescopes:** **Swope 1.0 m** (optical + NIR imaging) and **du Pont 2.5 m** (near-IR + spectroscopy).
- **Bands:** optical **u g r i B V** + near-IR **Y J H** (K also) → the **7-band ugriYJH** you cite.
- **CSP-I (2004-2009):** the gold standard of **well-calibrated, low- $z$  SN Ia** light curves. **CSP-II (2011-2015)** extended it.
- **Why NIR is the whole point (say this in viva):** 1. SN Ia are **better standard candles in the near-IR** (much less sensitive to dust reddening). 2. Host **NIR light traces old stars** → breaks the age-dust-metallicity degeneracy → **tighter host ages**.
- This is exactly why your **Paper 2** beats Son et al.'s **optical-only** age constraints.
- **Light-curve fitter:** **SNooPy** was built for CSP (handles NIR).

**One-line contrast for the panel:** *SDSS-II = many hosts at intermediate  $z$  with 5 optical bands; CSP-I = fewer but pristine low- $z$  SN with 7 optical+NIR bands, giving the age leverage my test needs.*

## SECTION 2 — Rubin / LSST (your PIE-ADA target — expect a question)

- **Vera C. Rubin Observatory**, Cerro Pachón, Chile. **Simonyi Survey Telescope, 8.4 m** primary.
- **LSSTCam = 3.2 gigapixel** — the **largest digital camera** ever built.
- **6 bands: u g r i z y**. Huge  **$9.6\text{ deg}^2$**  field of view.
- **LSST = Legacy Survey of Space and Time**, a **10-year** survey imaging the whole southern sky every few nights → **millions of transient alerts per night**. Commissioning/first-light **2025**, science operations ramping up **2025-2026**.
- **PLAsTiCC = Photometric LSST Astronomical Time-series Classification Challenge** — the simulated LSST dataset your **PIE-ADA** thesis trains on (14 transient classes, 6 passbands).
- **Viva bridge:** "My thesis prepares light-curve classification for the LSST data flood; my SN cosmology work is what you do with the classified SN Ia afterward."

## SECTION 3 — Other recent optical time-domain / SN surveys

| Survey                                 | Telescope                                                       | Bands              | What it's for                                                                               |
|----------------------------------------|-----------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------|
| <b>ZTF</b> (Zwicky Transient Facility) | Palomar <b>48-inch (1.2 m)</b> Schmidt, 47 deg <sup>2</sup> cam | g r i              | High-rate discovery of nearby transients; large modern SN Ia samples                        |
| <b>ATLAS</b>                           | Small wide-field 'scopes, all-sky                               | c, o (cyan/orange) | Early alerts; nearby SN & near-Earth objects                                                |
| <b>Pan-STARRS (PS1)</b>                | <b>1.8 m</b> , Haleakalā, Hawaii                                | g r i z y          | Medium-Deep Survey; <b>Foundation</b> low-z SN Ia sample                                    |
| <b>ASAS-SN</b>                         | Network of small telescopes                                     | V, g               | Catches <b>bright, nearby</b> SN across the whole sky                                       |
| <b>DES</b> (Dark Energy Survey)        | <b>Blanco 4 m</b> (CTIO), <b>DECam</b> 570 MP                   | g r i z Y          | <b>DES-SN 5-yr</b> : ~1500 photometric SN Ia; 2024 cosmology hinted at evolving dark energy |

**Point to make:** these are the surveys feeding the *next generation* of host-age / Hubble-residual studies — your method (Bayesian host SED fitting + sampler care) is what they will need at scale.

## SECTION 4 — Space telescopes (cosmology-relevant)

- **HST (Hubble, 2.4 m)**: high-z SN Ia and sharp **host-galaxy imaging**; long the backbone of the SN Ia Hubble diagram at high z.
- **JWST (6.5 m, infrared)**: highest-z SN, dust and old-population studies; refines the distance ladder.
- **Nancy Grace Roman Space Telescope (2.4 m, wide-field IR)**: launch ~**2026-27**; its **High-Latitude Time-Domain Survey** will find **thousands of SN Ia** for dark energy — *directly your future field; good to name*.
- **Euclid (ESA, 1.2 m, launched 2023)**: dark energy via **weak lensing + BAO** (a complementary probe, not SN).
- **Gaia (ESA)**: astrometry/parallax; also issues **Gaia transient alerts**.

## SECTION 5 — Radio facilities (for RGC + your I-HOW workshop)

- **VLA** (Karl G. Jansky Very Large Array): 27 dishes, New Mexico — classic radio-AGN imaging.
- **LOFAR** (Low Frequency Array): Europe-wide low-frequency array; **LoTSS** survey — the kind of radio-survey data behind **RGC**.
- **MeerKAT**: 64 dishes, South Africa — **SKA precursor**.
- **ASKAP**: Australian SKA Pathfinder; **EMU** all-sky continuum survey.
- **uGMRT**: upgraded Giant Metrewave Radio Telescope, India (low-frequency).
- **SKA** (Square Kilometre Array): under construction — **SKA-Low** (Australia) + **SKA-Mid** (South Africa); the future of radio astronomy (relevant to your **IAU Hands-on Workshop in Radio Astronomy**, Byurakan Sep 2026).
- Radio-AGN morphology recap: **FRI** (edge-darkened, core-bright, lower power) vs **FRII** (edge-brightened hotspots, higher power); emission = **synchrotron**.

## SECTION 6 — Spectroscopic / host-galaxy surveys (know the names)

- **DESI** (Dark Energy Spectroscopic Instrument): **Mayall 4 m**, Kitt Peak — millions of galaxy redshifts for **BAO**; the **2024 DESI** results (with DES/Union SN) hinted dark energy may be **evolving (w ≠ -1)** — a hot current topic; tie it to *why host systematics like yours matter now*.
- **SDSS MaNGA / SDSS-V**: spatially resolved (IFU) galaxy spectroscopy — relevant to **local vs global** host age (your global-age choice).
- **4MOST** (ESO): upcoming massive multi-object spectrograph.

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## Quick-recall (flash-card)

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- **SDSS:** 2.5 m, APO; **ugriz;** SDSS-II SN Survey → ~500 SN Ia,  $z \approx 0.05-0.4$ , Stripe 82 → **your Paper 1 hosts** (Rose et al. 2019, FSPS).
- **CSP:** Swope 1.0 m + du Pont 2.5 m, Las Campanas; **ugriYJH** (7-band, optical+NIR); low- $z$  gold sample → **your Paper 2**; NIR → tighter ages → beats Son et al.
- **Rubin/LSST:** 8.4 m, Chile; **ugrizy;** 3.2 GP camera; 10-yr survey; **PLAsTiCC = your PIE-ADA data.**
- **DES:** Blanco 4 m + DECam, **grizY**, ~1500 SN Ia.
- **ZTF:** Palomar 1.2 m, gri, high-rate discovery. **Pan-STARRS:** 1.8 m, grizy, Foundation sample.
- **Roman:** 2.4 m IR, ~2026-27, thousands of SN Ia — your future.
- **Radio:** VLA, LOFAR (LoTSS→RGC), MeerKAT/ASKAP (SKA precursors), SKA, uGMRT.
- **DESI:** Mayall 4 m, BAO; 2024 → hint of evolving dark energy.

## Likely viva Qs

1. Where does your Paper 1 / Paper 2 data come from, and what are the differences?
2. Why do you need near-IR (CSP) and not just optical (SDSS)?
3. What is LSST / Rubin, and how does your PIE-ADA relate to it?
4. Name a current-generation SN survey and what makes it powerful.
5. (RGC) Which radio facilities produce AGN survey data?